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To the Editor-in-Chief

Transportation Research Part B: Methodological

Submission: "Many small carriers: the mean-field limit of competitive urban delivery under zone-time charges"

Dear Editor,

I submit the above manuscript for consideration as a research article in Transportation Research Part B. It studies a city that charges delivery carriers by zone and hour when the carriers are many and small, and asks how quickly a finite market approaches its mean-field (Wardrop) limit and what that means for the charge the city should set.

The contributions are:

Rate tightness. For aggregative games with linear profits over routing plans, the Nash-Wardrop aggregate gap is of order the square root of the largest carrier share, and a two-zone family attains this order. Hence no bound of order one carrier's share with an instance-free constant exists once plans couple zones. The manuscript says plainly that the square-root rate and the O(1/M) rate under strong monotonicity in individual strategies are due to Paccagnan et al. (2019); the new part is the tightness family and the bounds where strong monotonicity fails.

Constant one. For carriers whose problems separate by zone-hour, the gap is at most one carrier's delivery, with equality for identical carriers; for replicas it is O(1/n) with a constant given by the conditioning of the carrier types.

Welfare bound. The city's observation-only charge, which uses only observed zone-hour totals, has a finite-market fixed point that equals a Nash aggregate of an auxiliary game, and its welfare loss is bounded uniformly in the instance.

City-scale evidence. On nine cities (with a stylised hourly profile, which the manuscript states and grades) the gap stays near one carrier's delivery from 24 to 1536 carriers. The tables show who gains and who loses: with few carriers the efficient charge is a subsidy that carriers and buyers welcome and the city pays; with many carriers all rules converge to the Pigou charge.

Entry and externalities. Lump-sum fees can fund the subsidy only with carrier-specific terms at small N; with free entry the subsidy attracts entrants and lowers welfare net of fixed costs; through-traffic delay, emissions and noise are several times the in-zone congestion charge.

The manuscript is 37 pages (A4, single column, 11 pt) with 52 references. The manuscript builds on a companion paper by the author, cited as Dang (2026) and under review. Code, instances, results and a Lean check of the algebraic cores are in the repository named in the data availability statement. The work has not been published elsewhere and is not under consideration by another journal.

Yours sincerely,

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